Trustworthy Machine Learning ICML 2024

Unveiling Privacy, Memorization, and Input Curvature Links

Deepak Ravikumar, Efstathia Soufleri, Abolfazl Hashemi, Kaushik Roy

OVERVIEW

Deep Neural Nets (DNNs) often overfit and memorize training data, impacting generalization, noisy learning, and privacy. Feldman (2019) proposed a formal memorization score, but it is computationally intensive. Recent work links input loss curvature with memorization, showing it to be much more efficient. We theoretically connect memorization, differential privacy, and input loss curvature, validating our findings on CIFAR and ImageNet datasets.

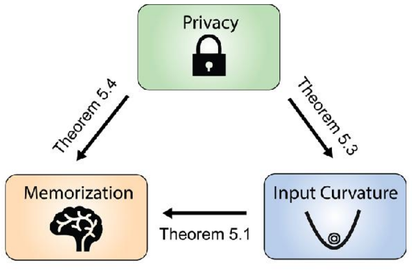
Motivation and Background

Low Input Curvature Typical Easy

Images from ImageNet ranked using input loss curvature. Input loss curvature was obtained using a single ResNet18 trained on ImageNet. Ten lowest curvature samples (top) and ten highest curvature samples (bottom) from the training set are visualized for 5 classes (each row is a class) from ImageNet. Low curvature samples are 'prototypical' of their class, while high curvature samples are rare, difficult, and more likely memorized instances

High Input Curvature Atypical Hard

Theory



Our theoretical framework provides upper bounds in Theorems 5.1, 5.3, and 5.4. These are visualized as links between Differential Privacy, Memorization, and Input Loss Curvature.

Theoretical Results

Input Curvature is given by

$$\mathrm{Curv}_{\phi}(z_i, S) = \mathrm{tr}(H) = \mathrm{tr}\left(\nabla^2_{z_i} l(h_S^{\phi}, z_i)\right)$$

Theorem 5.1 Input Curvature Upper Bounds Memorization

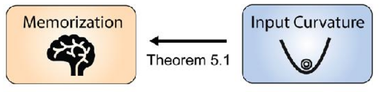
Lemma 5.2 Privacy ⇒ Stability

Theorem 5.3 Privacy ⇒ Low Input Loss Curvature

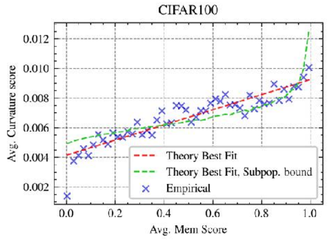
Theorem 5.4 Privacy ⇒ Less Memorization

Empirical Results

Memorization vs. Input Curvature

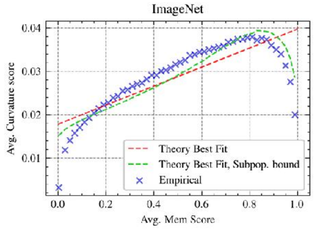


Here we aim to understand how memorization changes with curvature. The experiment aims to plot the memorization score vs. curvature score to validate our theoretical results. We calculate curvature scores by averaging over many seeds at the end of training. This measurement is proportional to the expected curvature score in Theorem 5.1

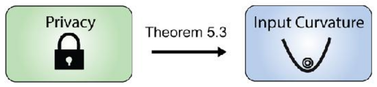


Plot of memorization score vs. input loss curvature at the end of training for CIFAR100 (average over 1000 Small Inception models)

Plot of memorization score vs. input loss curvature at the end of training for ImageNet (average over 100 ResNet50 models)

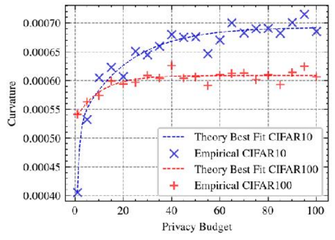


Differential Privacy vs. Input Curvature

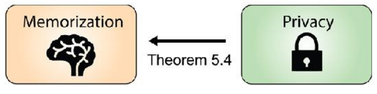


We study the relation between privacy and curvature, we train private ResNet18 models on CIFAR10 and CIFAR100 using DP-SGD (Abadi et al., 2016). We aim to plot privacy budget vs curvature score and validate Theorem 5.3

Result of studying the link between input loss curvature and privacy budget. All these results strongly correlate with theory and validate Theorem 5.3.

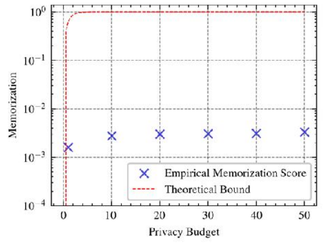


Differential Privacy vs. Memorization



We how privacy affects memorization for the top 500 most memorized examples. This is done by varying the privacy budget. The results align with Theorem 5.4, showing an increase in memorization as the privacy budget increases.

Plot of differential privacy vs memorization for CIFAR100 and the upper bound from the Theorem 5.4



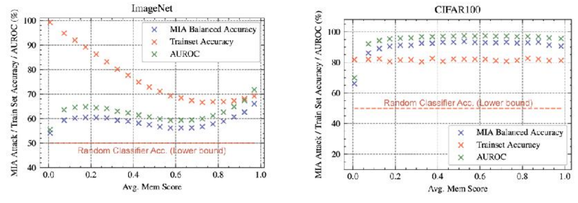
The memorization scores are significantly lower than the bound from Theorem 5.4 supporting Nasr et al. (2021)'s observation that DP-SGD may be overly conservative.

Summary

- We explore the theoretical link between memorization, curvature, and privacy.
- Theoretical analysis is based on assumptions of stability, generalization, and Lipschitzness, applicable to non-convex settings like DNNs.
- Main result: Memorization is upper-bounded by the curvature of the loss with respect to input and privacy.
- Privacy bounds input curvature and memorization
- Theory validated using DNNs on CIFAR100 and ImageNet, showing a strong match between theoretical predictions and empirical results.

MIA Perspective

A common approach to measure privacy is via membership inference attack (MIA), We used the LiRA (Carlini et al., 2022). More memorized samples are easier to detect using MIA attacks. For ImageNet MIA performance depends on the accuracy of the samples in the trainset. If the accuracy on the trainset is high and memorization is low, MIA is unsuccessful (mem. score < 0.1). If the memorization is high but the model is inaccurate (score range 0.4 – 0.7) then due to lack of model learning MIA is unsuccessful but to a lesser extent. However, if memorization is high and accuracy on the samples is high MIA is successful (mem. score > 0.7).



References

Abadi, M., Chu, A., Goodfellow, I., McMahan, H. B., Mironov, I., Talwar, K., and Zhang, L. Deep learning with differential privacy. In Proceedings of the 2016 ACM SIGSAC conference on computer and communications security, pp. 308–318, 2016

Feldman, V. Does learning require memorization? a short tale about a long tail. arXiv preprint arXiv:1906.05271, 2019

Nasr, M., Songi, S., Thakurta, A., Papernot, N., and Carlini, N. Adversary instantiation: Lower bounds for differentially private machine learning. In 2021 IEEE Symposium on security and privacy (SP), pp. 866–882. IEEE, 2021.